

Sungguk Cha

Biography

I am Researcher at the Service AI Research Team of LG Uplus closely working with Dr. Mintae Kim and DongWook Kim. Before that, I have spent wonderful times as a Technical Research Personnel for the military duty working with Dr. Cheoljong Yang and Jusung Lee at NCSoft, and a number of fantastic colleagues at MINDs Lab Inc. I obtained my M.S. degree in Ulsan National Institute of Science and Technology.

My research interests cover the broad area of machine learning, multi-modal learning, computer vision, language modeling and quantitative finance. My current position focuses on synthesizing train/evaluate data for (vision-)language models.

Publications

- Frame Entrepreneurs in an AI Agent Community: Concentrated Identity-Claim Production on Moltbook
Sungguk Cha, DongWook Kim.
Preprint, 2026.
- ReinPool: Reinforcement Learning Pooling Multi-Vector Embeddings for Retrieval System
Sungguk Cha, DongWook Kim, Mintae Kim, Youngsub Han, Byoung-Ki Jeon, Sangyeob Lee
Preprint, 2026.
- Annotation-Free Reinforcement Learning Query Rewriting via Verifiable Search Reward
Sungguk Cha, DongWook Kim, Taeseung Hahn, Mintae Kim, Youngsub Han, Byoung-Ki Jeon
Preprint, 2025.
- Visually Dehallucinative Instruction Generation: Know What You Don't Know
Sungguk Cha, Jusung Lee, Younghyun Lee, Cheoljong Yang.
Preprint, 2024.
- Visually Dehallucinative Instruction Generation
Sungguk Cha, Jusung Lee, Younghyun Lee, Cheoljong Yang.
Proceedings of IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2024.
- Question Answer Instruction: Unlocking Multi-Modal Large Language Model to Domain-Specific Visual Multi-Tasks
Jusung Lee, **Sungguk Cha**, Younghyun Lee, Cheoljong Yang.
Preprint, 2024.
- Zero-Shot Semantic Segmentation via Spatial and Multi-scale aware Visual Class Embedding
Sungguk Cha*, Yooseung Wang*.
Pattern Recognition Letters (PRL), Volume 158, June 2022, Pages 87-93.

Acknowledgements

- Top Finisher in StarCraft AI Tournaments
Entry name: Zia Bot
 - StarCraft AI Competitions, Bots and Tournament Manager Software.
Michal Čertický, David Churchill, Kyung-Joong Kim, Martin Čertický, Richard Kelly.
IEEE Transactions on Games (ToG): 1-13. DOI: 10.1109/TG.2018.2883499. Print ISSN: 2475-1502. Online ISSN: 2475-1510. 2018.
 - The Current State of StarCraft AI Competitions and Bots.
M. Čertický, D. Churchill.
In Proceedings of the AIIDE 2017 Workshop on Artificial Intelligence for Strategy Games. 2017.

Experiences

LG Uplus

August 2024 - Present

- Service AI Research Team (Jan. 2026 - Present)
 - Reference: Dr. Mintae Kim
 - Multimodal Embedding alignment and Interleaved Data Processing.
 - GUI Agent.
- RAG AI Technical Team (Jan. 2025 - Dec. 2025)
 - Reference: Dr. Mintae Kim
 - Multimodal RAG, document question answering
 - Multimodal RAG framework implementation including document embedding and question answering.
 - Multi-turn RAG agent developments and evaluation dealing with the company secretary documents. (reference Geun Lim)
 - Built the train/evaluate data for the multiturn query rewrite agent.
 - Built the evaluate data and the llm-as-judge model for the multiturn RAG scenario.
 - Closely worked with the related departments preparing benchmarks, data and handling issues related to the RAG system.
 - Document Parsing (reference Soyeon Jo, Heonjin Ha)
 - Advancing document parsing (Document Layout Analysis(DLA), VLM parsing) : analysis, evaluation and synthesizing.
 - Query Rewriting (reference DongWook Kim, Taeseung Hahn)
 - (Data, Index)-agnostic query rewriting approach for retrieval system.
- LLM Technical Team (Aug. 2024 - Dec. 2024)
 - Developed multi-turn RAG data synthesis approaches.
 - MWC 2025 preparation. LLM for digital humanoid implementation.

OAQ Co. Ltd.

June 2024 - March 2025

- Auditor

NCSoft

Sep. 2022 - May 2024

- Technical Research Personnel for Alternative Military Service (Sep. 2022 - Feb. 2024)
- Image Generation Team, Multimodal AI Lab. (Feb. 2024 - May 2024)
 - VQA (VLM) research and development for digital humans.
 - Multi-modal model research and development for game-domain. It includes multi-modal crawling webscale data.
 - Scale-up Data Curation for Generative Foundation Models: Replaced human-annotation in captioning (tagging) and resulted in dramatic improvements in developing image generative models.
- Recognition Team, Vision AI Lab., NC Research (Sep. 2022 - Feb. 2024)
 - Multimodal Recognition AI: Generated multi-modal data for novel tasks. Researched on evaluating large multimodal models. Responsible and dehallucinative multimodal data generation.
 - Conversational AI
 - Facial Expression Recognition

MINDs Lab., Inc.

Sep. 2020 - Sep. 2022

- The company now goes by the name maum.ai.
- Technical Research Personnel for Alternative Military Service (Feb. 2021 - Sep. 2022)
- Senior AI Scientist (Mar. 2021 - Sep. 2022)
 - Co-led the research team (HR, managing, etc.)

- Developed and implemented fundamental visual recognition applications, focusing on classification such as firearms and OCR, detection including mover objects and car damage, segmentation with a specialization in fashion items, and person re-identification for tracking corona-infected individuals.
- Contributed to Conversational AI, specializing in human-object interaction through achievements in Visual Question Answering, ATM abnormal behavior detection, and human-object detection/classification.

Statistical Artificial Intelligence Lab

Jul. 2018 - Feb. 2019

- <http://sailab.kaist.ac.kr/>
- Adviser: Prof. Jaesik Choi
- Undergraduate Researcher
- Research Topic: Drivable Area Semantic Segmentation

Agents and Robotic Transportation Lab

Dec. 2017 - Feb. 2018

- Adviser: Prof. Tsz-Chiu Au
- Undergraduate Researcher
- Research Topic: StarCraft AI Algorithm

Machine Learning & Artificial Intelligence

Sep. 2017 - Dec 2017

- <https://www.mlai-kaist.com/>
- Adviser: Prof. Sung Ju Hwang
- Undergraduate Researcher
- Research Topic: Variational Auto Encoder for Text

SKT T-Brain

Mar. 2017

- Research service invited by Hyunsoo Kim (Superb AI CEO)
- Passed interviews
- Research Topic: Machine Learning Application for StarCraft

Professional Services

Conference Talks:

- Frame Entrepreneurs in an AI Agent Community: Concentrated Identity-Claim Production on Moltbook
Sungguk Cha, DongWook Kim.
Communication, Information Technologies, and Media Sociology section of the American Sociological Association (CITAMS), 2026.
- 2026 American Sociological Association (ASA) Annual Meeting CITAMS Roundtable: "The Synthetic Social Graph: Emergent Behavior in AI Agent Communities"
- LG Software Developer Conference (LG SDC) 2025.

Conference Reviewer:

- Association for the Advancement of Artificial Intelligence (AAAI) 2026.
- IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) 2025, 2026.
- IEEE International Joint Conference on Neural Networks (IJCNN) 2025, 2026.

Educations

Ulsan National Institute of Science and Technology

Mar. 2019 - Feb. 2021

- **Master of Science** in Computer Science and Engineering (GPA 3.94/4.30)
- Thesis: Zero-Shot Semantic Segmentation via Visual Class Embedding
- Full Merit based Scholarship

Ulsan National Institute of Science and Technology

Mar. 2016 - Feb. 2019
Mar. 2014 - Feb. 2015

- **Bachelor of Science** in
 - Computer Science and Engineering (GPA 3.74/4.30)
 - Electrical Engineering
- **Cum Laude**
- Full Merit based Scholarship

Honors & Awards

- Dacon AI Bit Trader Competition **29th (Top 21%)** Mar 2021
- NAVER LABS Mapping & Localization Challenge 8th place Jun 2020
- Kaggle/iMaterialist (Fashion) 2019 at FGVC6/fashion item panoptic segmentation/(29th/194 (Top 15%))
- Kaggle/Google Analytics Customer Revenue Prediction/(570th/3611 (Top16%))
- Student StarCraft AI Tournament **4th** Feb 2018
- Naver UNIST Undergraduate Poster Award, **Bronze** Prize Dec 2017
Topic: Reinforcement Learning in StarCraft in Real-Time using Deep Learning
- Samsung Dream Class Mentoring Scholarship Spring 2017
- AAAI AIIDE StarCraft AI Competition, 28th 2017
- IEEE CIG StarCraft AI Competition **3rd** 2017
- Student StarCraft AI Tournament **3rd** Feb 2017
- AAAI AIIDE StarCraft AI Competition, 11th 2016
- IEEE CIG StarCraft AI Competition 10th 2016
- Korean Olympiad of Informatics **Encouragement** Prize 2013
Regional representative of Daejeon
- Daejeon Olympiad of Informatics **Gold Prize** 2013
- Korean Olympiad of Informatics **Encouragement** Prize 2012
Regional representative of Daejeon
- Daejeon Olympiad of Informatics **Gold Prize** 2012
- Korean Olympiad of Informatics **Encouragement** Prize 2010
Regional representative of Daejeon
- Daejeon Olympiad of Informatics **Gold Prize** 2010

Teaching

- [TA] UNIST/CSE241: Object Oriented Programming Spring 2020
- [TA] UNIST/CSE251: System Programming Fall 2019
- **[Tutor] PyCon Korea** Aug 2019
Topic: Explainable AI
- [TA] UNIST/ITP117: Introduction to AI Programming II Spring 2019
- [TA] UNIST/AHS161: What is "I" Spring, Fall 2017

Skills

- Languages
 - **Korean** | **English** | **Python** | C/C++ | MFC

- Frameworks
 - Deep learning: **PyTorch, Multimodal, RLHF/RLAIF, RAG agents**
 - Collaboration: **ATLASSIAN**, Tigris
 - Developments: **Data Curation, Interleaved Data Processing**, Git, Docker, gRPC